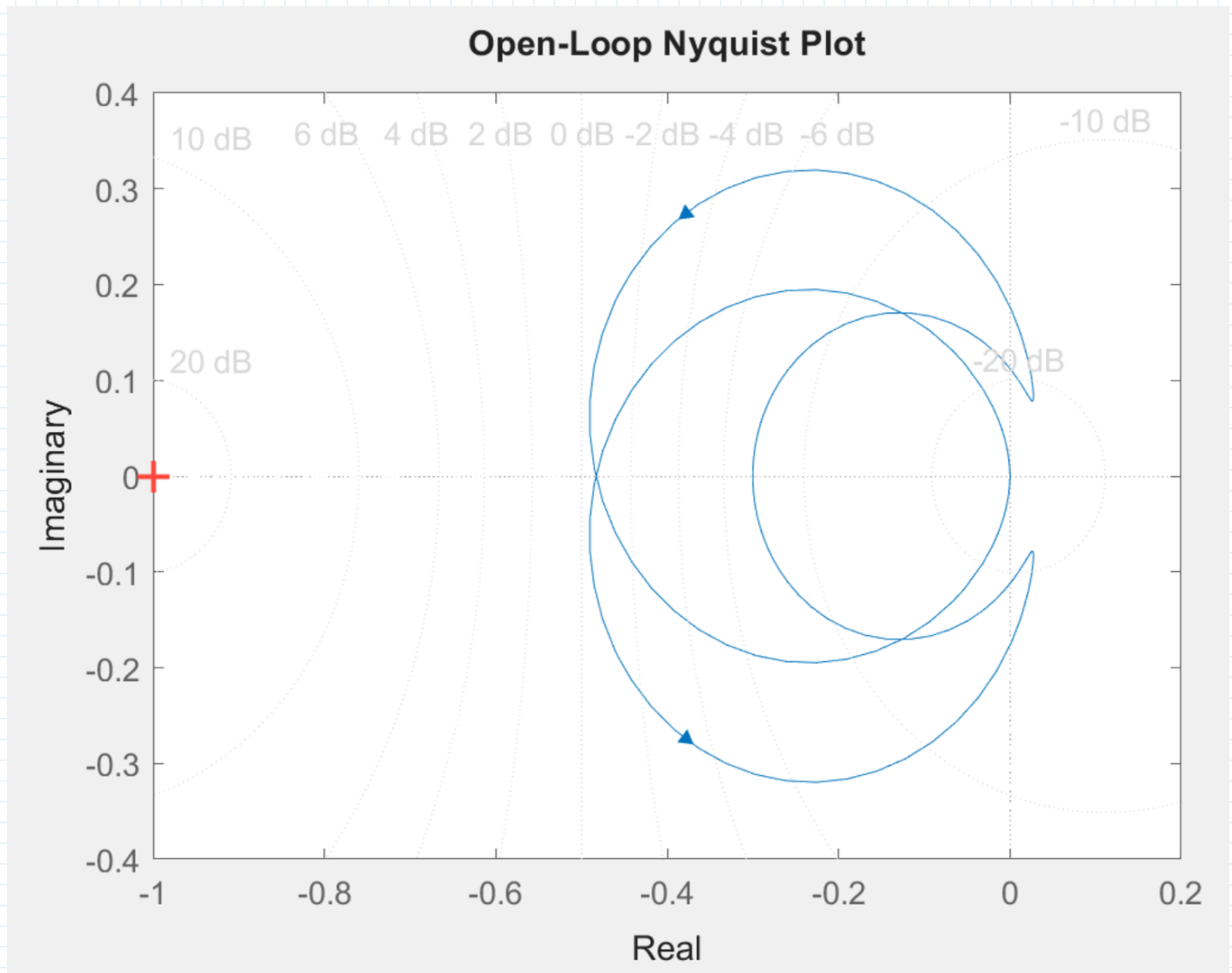


Recap

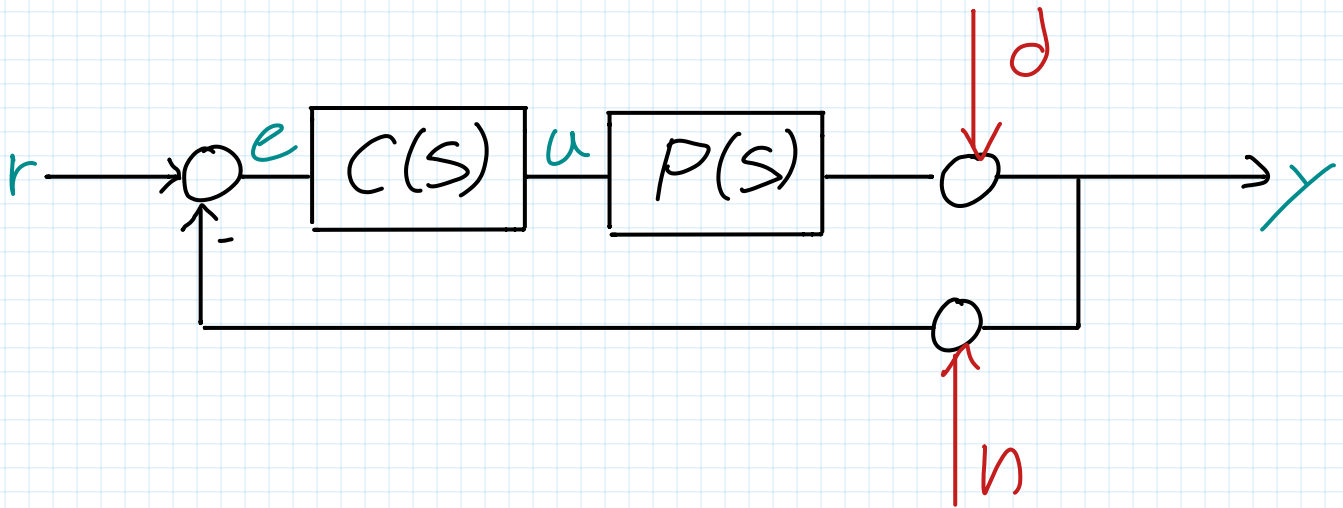
- Last week we introduced the Nyquist plot. With it, we were able to evaluate **closed**-loop stability, only by looking at the **open**-loop tf.



- We saw that we not only want to be stable, but also robust!

$\Rightarrow$ Stable even in the presence of uncertainties and unwanted inputs

Let's take a closer look at how exactly our system can be affected:



The effect of n and d is seen in the tf's from:

• $d \rightarrow e$:
$$S(s) = \frac{1}{1 + C(s)P(s)}$$

• $n \rightarrow e$:
$$T(s) = \frac{C(s)P(s)}{1 + C(s)P(s)}$$

$\Rightarrow$ The effect of both n and d on e should be as small as possible.

- That means, both $S(s)$ and $T(s)$ should be as small as possible.

Theres one small issue...

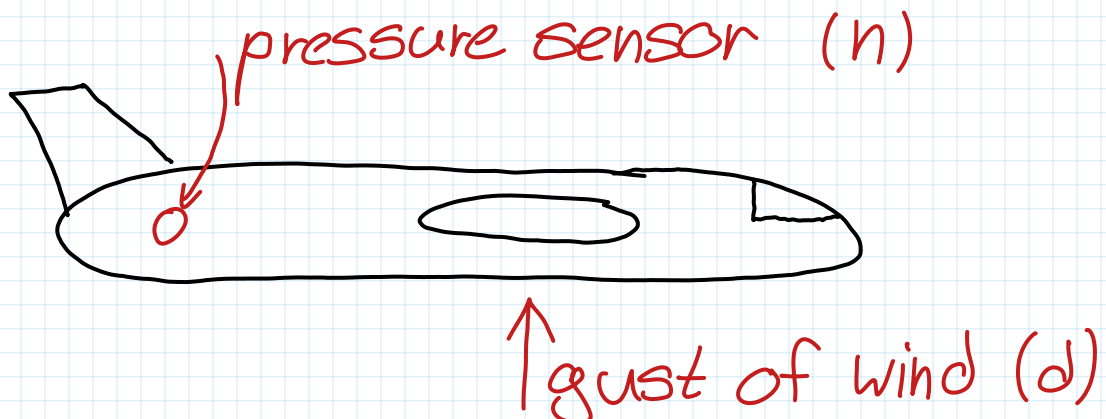
$$S(s) + T(s) = \frac{1}{1 + C(s)P(s)} + \frac{C(s)P(s)}{1 + C(s)P(s)} = 1$$

$\Rightarrow$ They always add up to one!

- This means that we cant cancel both noise and disturbances...

Thankfully they have a property we can make use of:

- Noise usually has a high frequency
- Disturbances usually have a low frequency.



• So the workaround is to require

1) $|S(s)| \ll 1$ for small ω
 $\Rightarrow$ Disturbance rejection

2) $|T(s)| \ll 1$ for large ω
 $\Rightarrow$ Noise rejection

How can we guarantee these constraints?

1) $S(s) = \frac{1}{1 + L(s)} \Rightarrow S(s)$ small, at small ω .

Now we enforce the following inequality, using a constraint function $W_1(s)$

$$|S(s)| \cdot |W_1(s)| < 1$$

For some $W_1(s)$, that is large at low ω .

The equation can be approximated as

$$\left| \frac{1}{1 + L(s)} \right| \cdot |W_1(s)| < 1 \Rightarrow |1 + L(s)| > |W_1(s)|$$

$$\approx |L(s)| > |W_1(s)| \quad (\text{for small } \omega)$$

$$2) T(s) = \frac{L(s)}{1 + L(s)} \Rightarrow T(s) \text{ small at high } \omega$$

Again using a constraint function:

$$|T(s)| \cdot |W_2(s)| < 1$$

For some $W_2(s)$ that is large at high ω

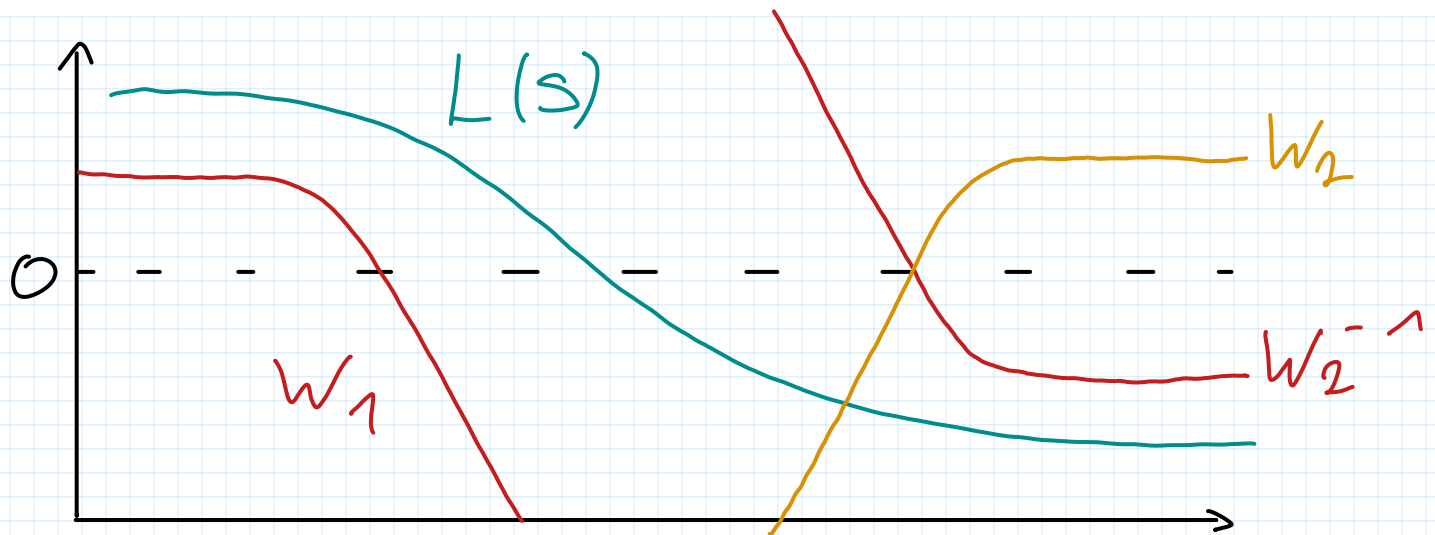
• Again we rewrite and approximate:

$$\left| \frac{L(s)}{1 + L(s)} \right| |W_2(s)| < 1 \approx |L(s)| |W_2(s)| < 1 \quad \text{for high } \omega$$

$$\Rightarrow |L(s)| < |W_2(s)|^{-1}$$

What exactly are these $W(s)$??

Their impact can be observed as "obstacles" in the Bode plot as such:



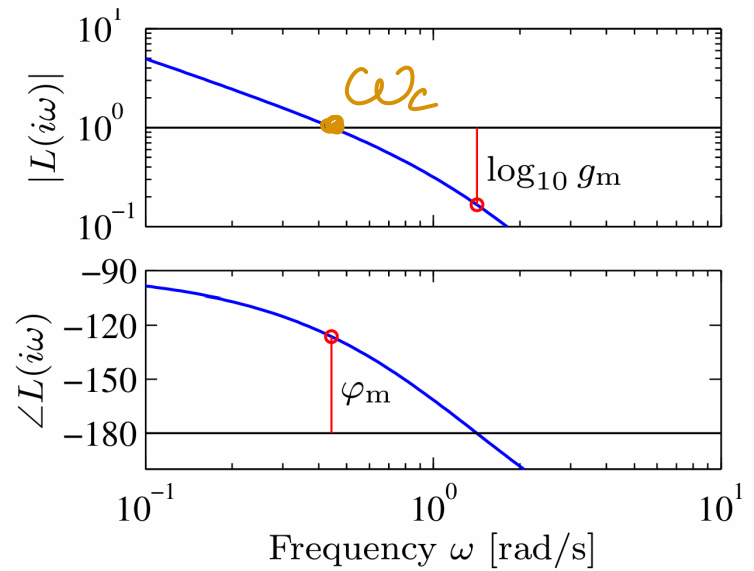
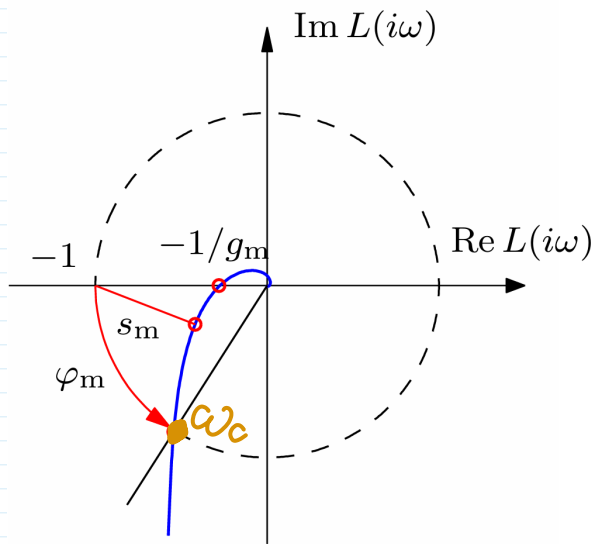
• The W 's are like obstacles for the open-loop tf, that *we can design*.

3) There's one more important condition for the closed loop:

$$T(s) = \frac{L(s)}{1 + L(s)}$$

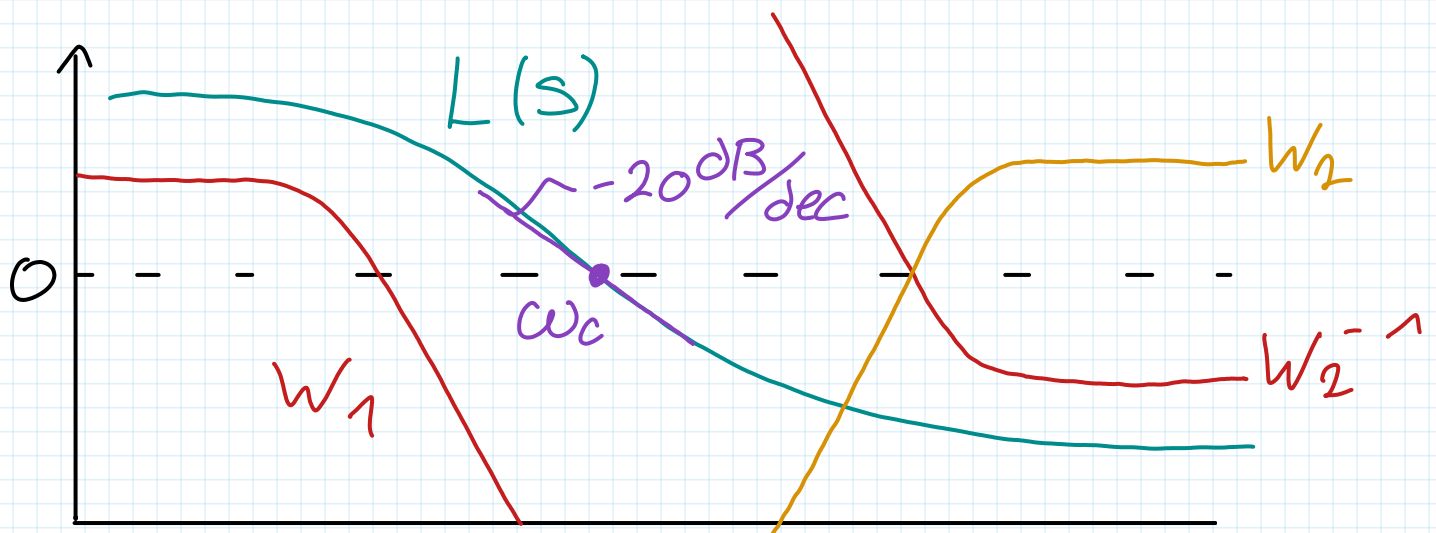
$\Rightarrow$ system becomes unstable for $L(s) = -1$.

$\Rightarrow$ We could also see this in the Bode-plot!



The slope of the magnitude plot is related to the phase!

- To ensure enough phase margin, the slope of the magnitude plot at the **crossover frequency** ω_c , should be at least -20dB/dec or flatter.



4) The condition at the crossover frequency ω_c is now established, but how do we even choose ω_c itself?

$\Rightarrow$ We must take a little detour..

Bandwidth ω_b

- Maximum frequency at which closed-loop can track references with a maximal attenuation of $1/\sqrt{2} \approx 0.7$

$\Rightarrow \omega_b \uparrow$: System reacts easily to fast signals

$\Rightarrow \omega_b \downarrow$: System struggles to follow even slow signals

Choose ω_b , s.t. that maximum signal speed you need to follow is still included.

From Bandwidth ω_b to crossover ω_c

- ω_b mathematically:

$$\omega \text{ s.t. } |T(j\omega)| = \frac{|L(j\omega)|}{|1+L(j\omega)|} = \frac{1}{\sqrt{2}}$$

which is true for $\angle L(j\omega) = \pm 90^\circ$

$$\begin{aligned} \angle(j\omega_b) &\Rightarrow |L(j\omega_b)| = 1 = 0\text{dB} = \omega_c \\ \angle L(j\omega_b) &= \pm 90^\circ \end{aligned}$$

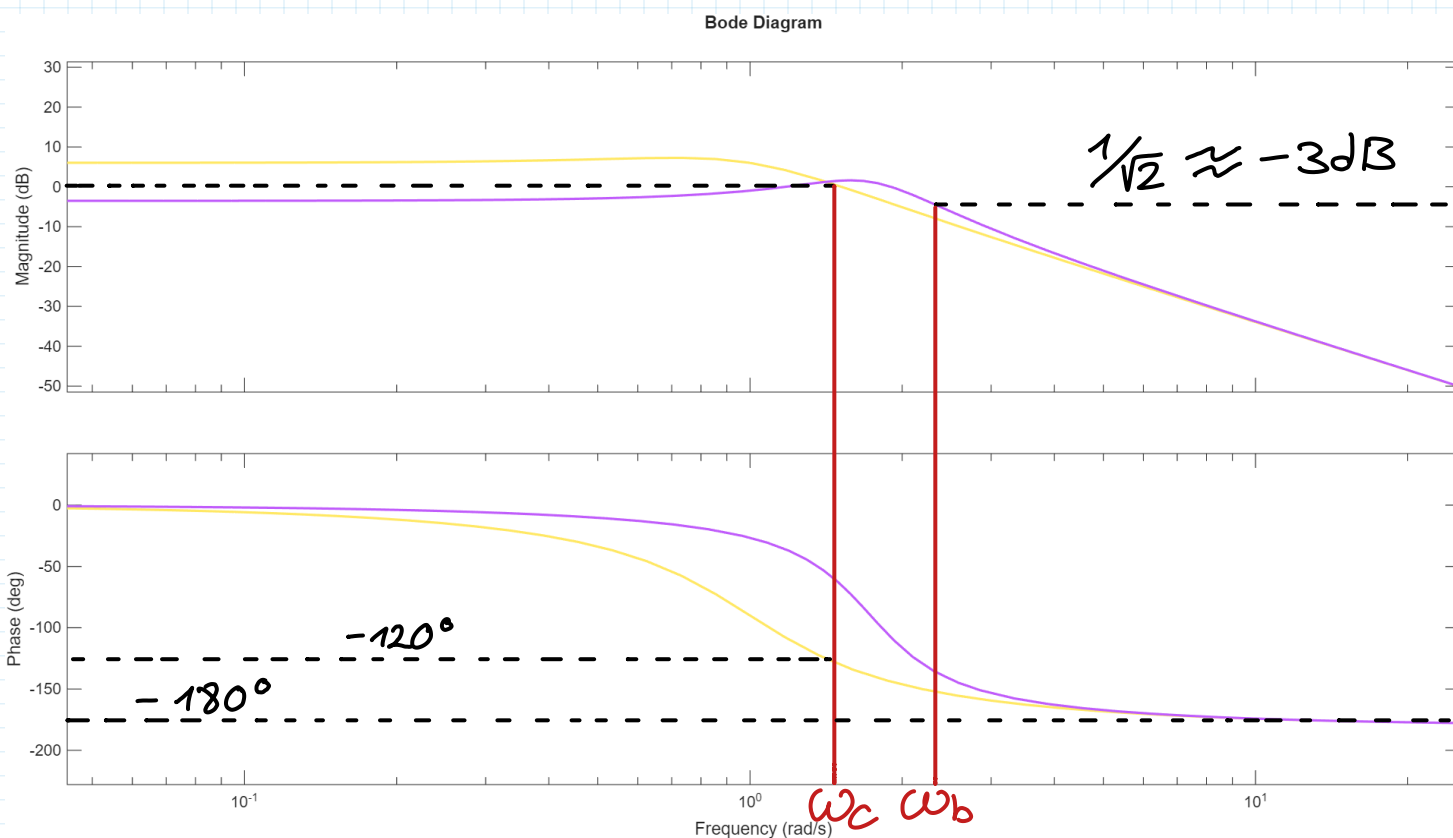
$\Rightarrow$ With a phase margin of 90° , $\omega_b = \omega_c$

- Since we anyways desire $\varphi_m \approx 80^\circ$, we choose the open-loop ω_c such that the closed-loop requirement for ω_b is met.

Example

Pendulum $P(s) = \frac{2}{s^2 + s + 1}$ (linearized at $(0,0)$)

and closed-loop $T(s) = \frac{P(s)}{1 + P(s)}$ (unit feedback)



$$\varphi_m = -180^\circ - (-120^\circ) = -60^\circ \neq -80^\circ$$

$\Rightarrow$ Hence $\omega_c \approx \omega_b$ and not $\omega_c = \omega_b$

We have successfully formulated our closed-loop conditions solely on the open-loop $L(s)$.

So if we have a plant $P(s)$ given, how do we now ensure the right frequency dependent behaviour of $L(s)$?

- $L(s) = k \cdot P(s)$ isn't going to work anymore...
 $\Rightarrow$ We will need a dynamic compensator $C(s)$!

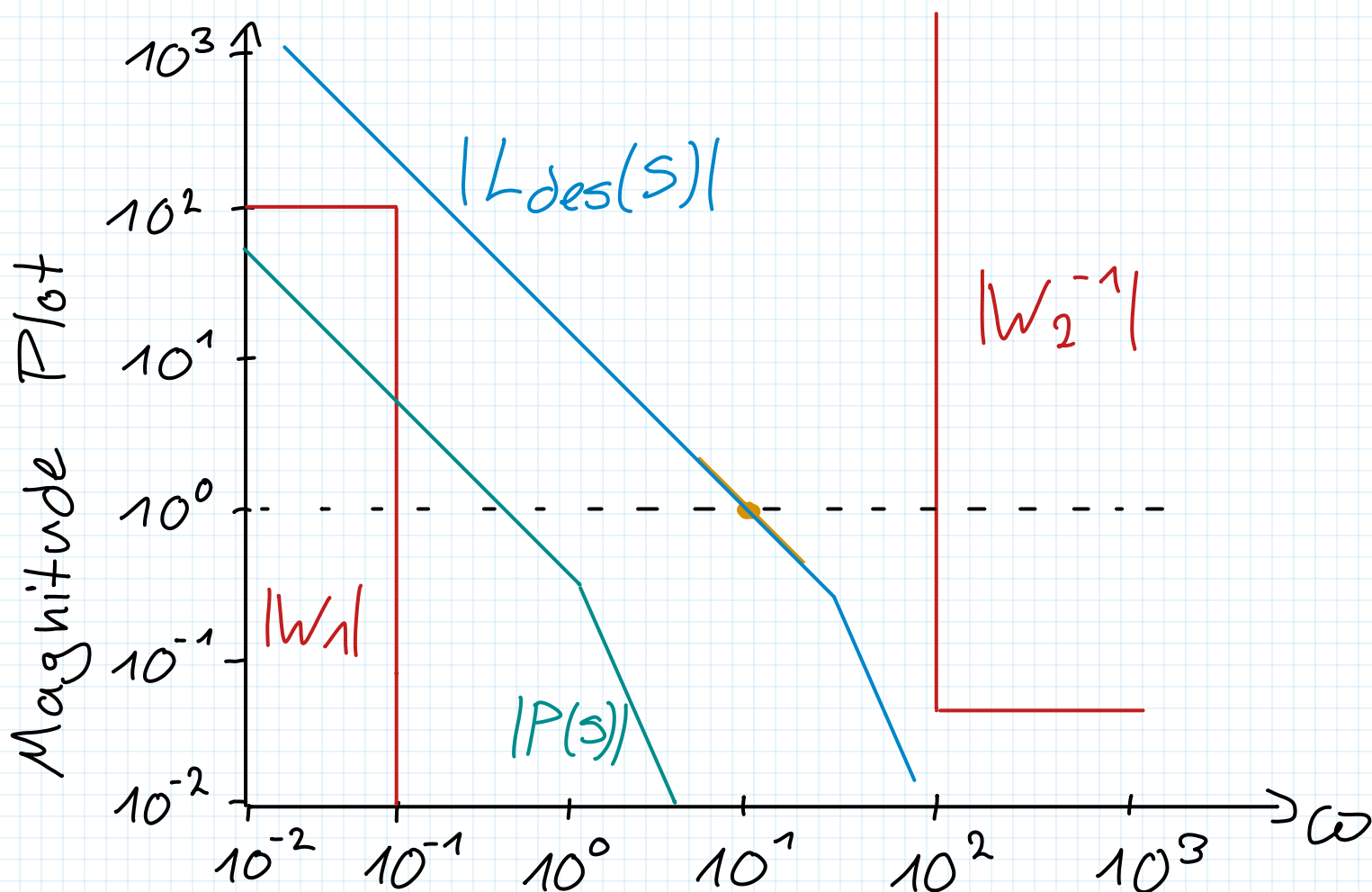
$$L(s) = C(s)P(s)$$

$\Rightarrow$ Choose the controller s.t. $L(s)$ does what we want!

Ex: Given $P(s) = 0,01 \cdot \frac{(s+10)^2}{s(s^2+2s+2)}$

Find a $C(s)$, s.t. the following conditions are respected:

- $W_1 = 100\text{dB}$ for $\omega < 0,1\text{rad/s}$
- $W_2^{-1} = 0,05\text{dB}$ for $\omega > 100\text{rad/s}$
- $\omega_c = 10\text{rad/s}$
- 45° Phase margin at ω_c



- A good open-loop suggestion would be:

$$L_{des} = 10 \cdot \frac{1}{s(s/50 + 1)}$$

- Now we reverse engineer:

$$C(s) = \frac{L_{des}(s)}{P(s)} = 10 \cdot \frac{s^2 + 2s + 2}{(s/50 + 1)(s/10 + 1)^2}$$

What we just did was a "brute force" approach...

- $C(s)$ may be unnecessarily complicated
 - Only works for open-loop stable, minimum-phase systems!
 - A more sophisticated approach is called loop-shaping
- $\Rightarrow$ Design a controller with the following elements, s.t. $L(s)$ passes the obstacle course:

- Gain: k
- Integrator: $1/s$
- Lead: $\frac{s/a + 1}{s/b + 1}$, $0 < a < b$
- Lag: $\frac{s/a + 1}{s/b + 1}$, $0 < b < a$

$\Rightarrow$ These can be combined however we want. But is the effect of Lead/Lag-elements?

Lead-Lag-Elements

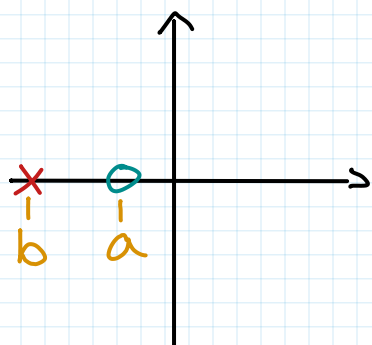
- In essence, Lead-Lag elements are a combination of a stable pole and a minimum-phase zero.

We use them to manipulate a frequency response

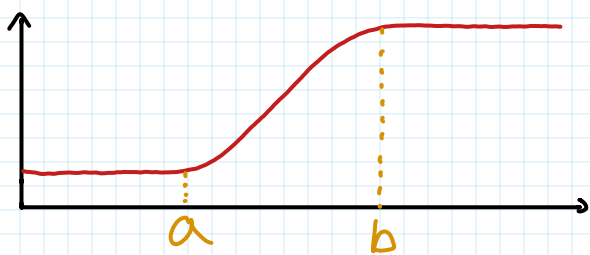
Lead

$$C_{\text{lead}} = \frac{s/a + 1}{s/b + 1}, \quad 0 < a < b$$

• Roots:



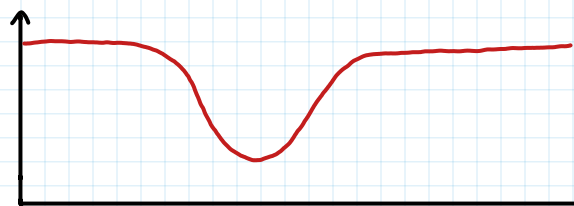
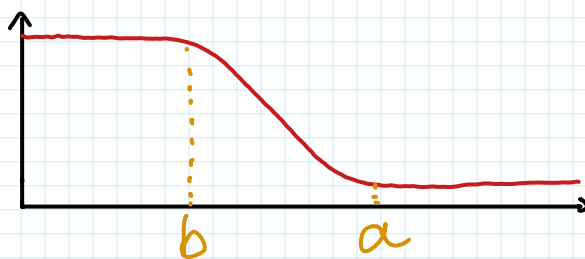
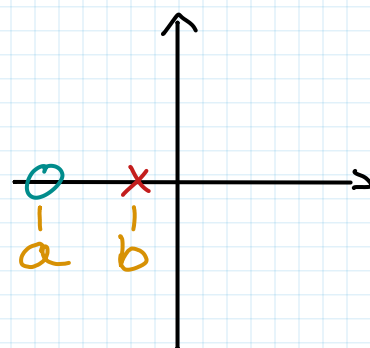
• Bode:



Lag

$$C_{\text{lag}} = \frac{s/a + 1}{s/b + 1}, \quad 0 < b < a$$

• Roots:



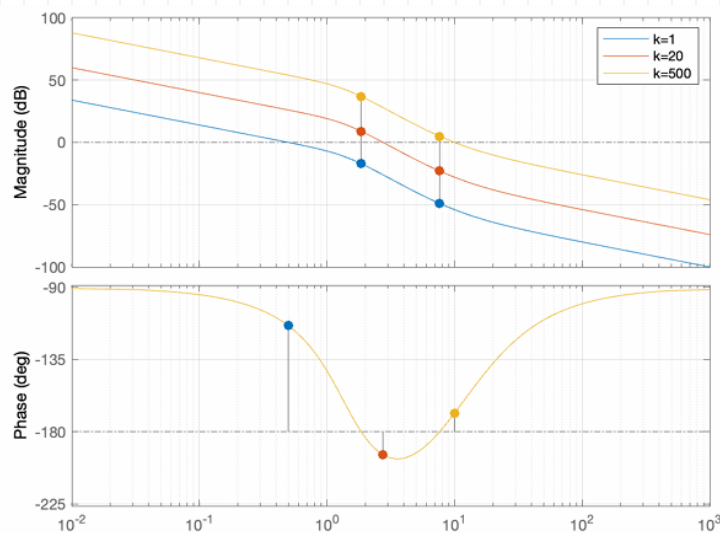
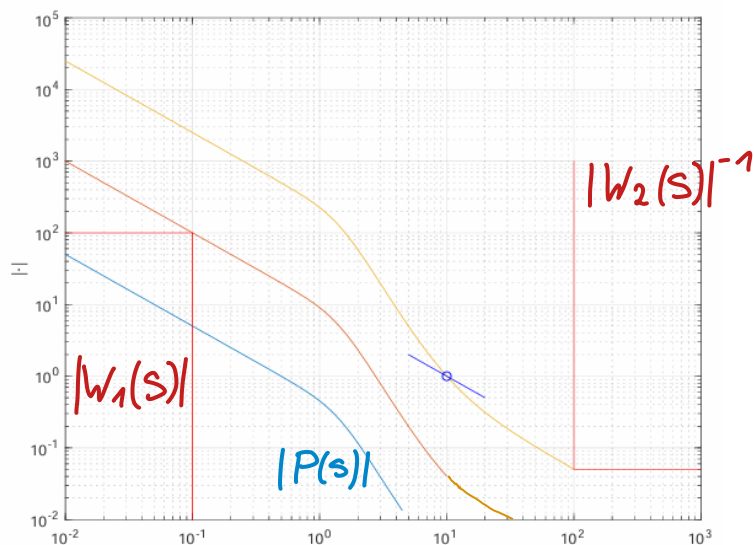
- Increase magnitude by: b/a at high ω
- Increase phase around $\omega = \sqrt{ab}$, with max. increase:

$$\varphi_{\max} = 2 \tan^{-1} \left(\sqrt{\frac{b}{a}} \right) - 90^\circ$$
- Decrease magnitude by: b/a at high ω
- Decrease phase around: $\omega = \sqrt{ab}$, with min. decrease:

$$\varphi_{\min} = 2 \tan^{-1} \left(\sqrt{\frac{b}{a}} \right) - 90^\circ$$

$\Rightarrow$ We can now build a controller brick by brick

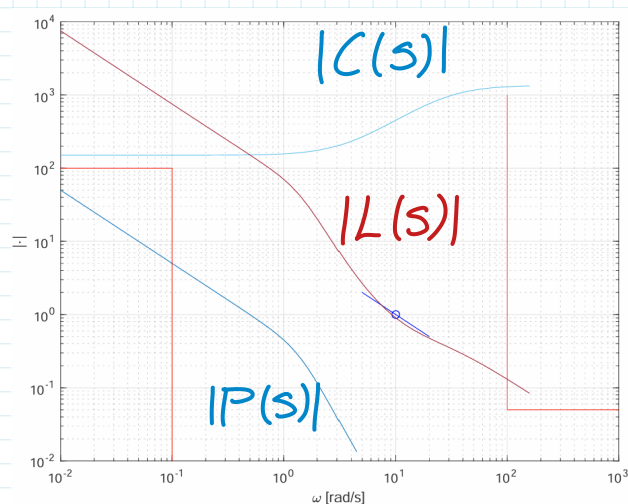
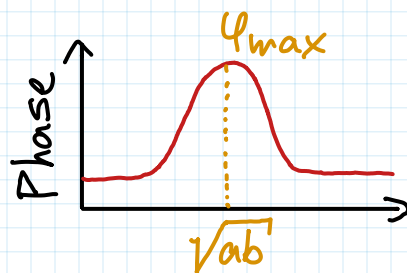
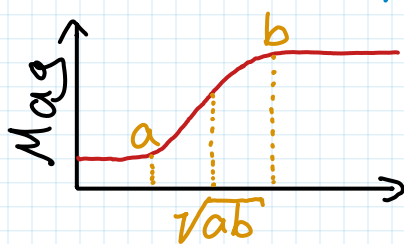
- First, look at the effect of k .



- $k=1$: No disturbance rejection
- $k=20$: Encircling -1 point $\Rightarrow$ unstable
- $k=500$: Okay but tiny phase margin!

Let's settle for $k=150$. How do we address the missing phase margin and insufficient slope at the crossover frequency?

=> Add a lead element!



• After rescaling k , we get:

$$C(s) = \underbrace{150}_K \underbrace{\frac{s/3.3 + 1}{s/30 + 1}}_{\text{lead}}$$

How did we choose a, b ?

• Pick $\sqrt{ab}$ at ω_c

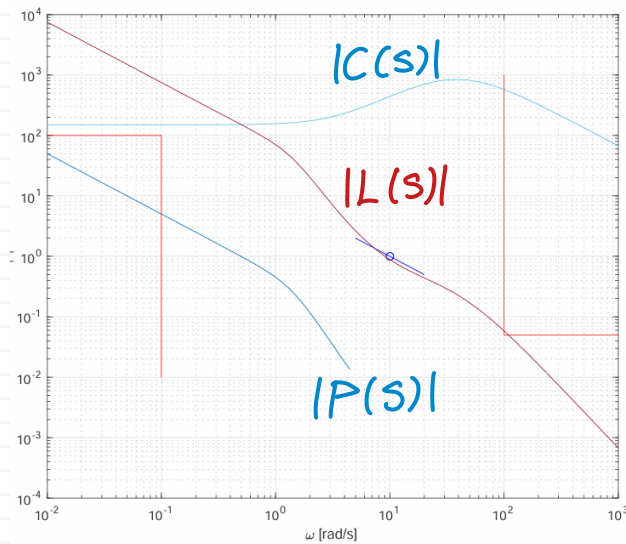
$$\Rightarrow \sqrt{ab} = 10^1 \text{ rad/s}$$

• Pick $\frac{b}{a}$ by how much $\phi_{\max}$ should be:

$$\phi_{\max} = 2 \arctan\left(\sqrt{\frac{b}{a}}\right) - 90^\circ$$

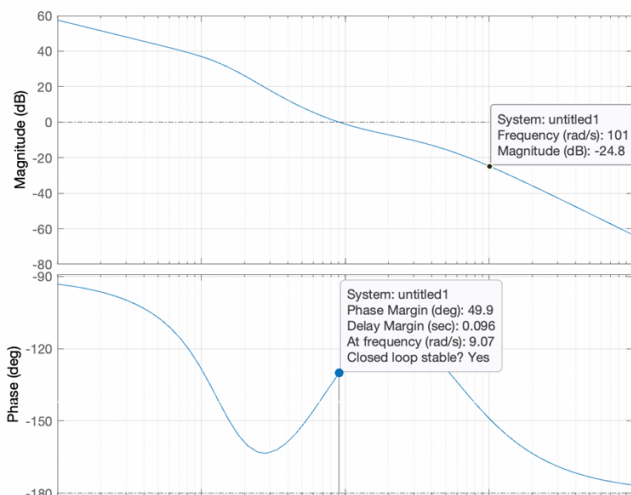
One problem left... We also need to reject noise... What element could we use for this?

$\Rightarrow$ Add a lag element!



$$C(s) = \underbrace{150}_k \cdot \underbrace{\frac{s/3.3 + 1}{s/30 + 1}}_{\text{lead}} \cdot \underbrace{\frac{s/1000 + 1}{s/50 + 1}}_{\text{lag}}$$

$\Rightarrow$ Now $L(s)$ fulfills all demands!



Important note: All of the above works only for open-loop stable systems...

It seems as though we can just do what we want with our open-loop right?

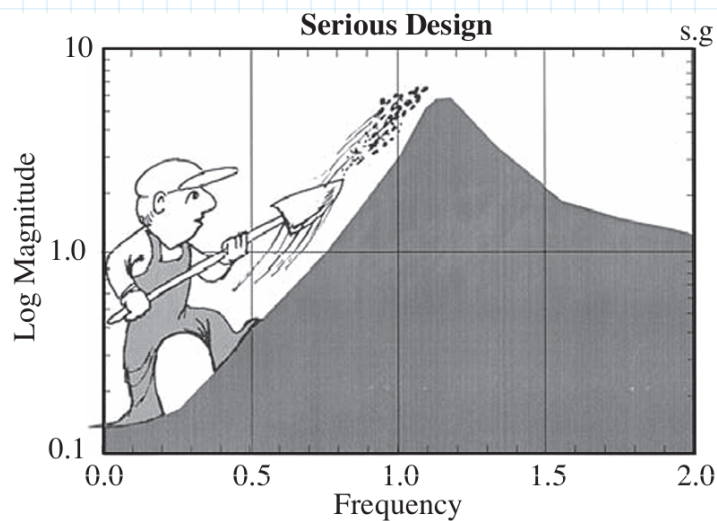
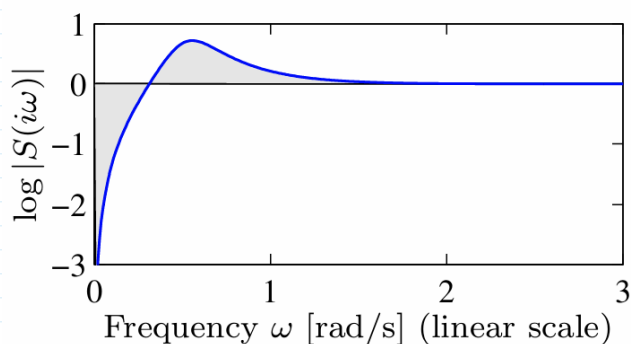
Bode's Integral

Let $S(s)$ be the sensitivity function of a stable closed-loop. Then:

$$\int_0^{\infty} \log(|S(j\omega)|) d\omega = \pi \sum p_k,$$

where p_k are the unstable poles of $L(s)$.

What does this mean?



Area underneath $S(s)$ remains the same!

Let's look at an exam question:

Problem: Consider a lead compensator

$$C_{\text{lead}}(s) = \frac{s/a + 1}{s/b + 1}, \quad 0 < a < b, \quad (1)$$

used in a standard feedback architecture to control a given plant P .

Q37 (0.5 Points) Mark the correct answer for each statement.

Statement	True	False
A lead compensator is typically used to reduce the phase margin.		☒
A lead compensator can increase the sensitivity to high-frequency noise.	☒	
The slope of the magnitude at frequencies between a and b is approximately +20 dB/decade.	☒	

Q38 (1.5 Points) A lead compensator as given in Equation (1) is to be used in order to **increase the phase** of the open-loop system $P(s) \cdot C_{\text{lead}}(s)$ by at most **30°** at a frequency of **10 rad/s**.

Which choice of the parameters a and b satisfies the aforementioned requirement?
Mark the correct answer.

- ☒ $a = \frac{10}{\sqrt{3}}, b = 10\sqrt{3} \rightarrow \omega = \sqrt{100} = 10$
- ☐ ~~$a = 1\sqrt{3}, b = 100\sqrt{3} \rightarrow \omega = \sqrt{300} \neq 10$~~
- ☐ $a = 10\sqrt{3}, b = \frac{10}{\sqrt{3}} \rightarrow \omega = \sqrt{100} = 10$
- ☐ ~~$a = 100\sqrt{3}, b = 1\sqrt{3} \rightarrow \omega = \sqrt{300} \neq 10$~~

$$\omega = \sqrt{ab}$$

lead: $b > a$

$\Rightarrow \textcircled{A}$

Q1) W_1 & W_2 make sure, that the closed loop is stable

☐ TRUE ☒ FALSE

=> Performance criteria!

Q2) In general, the loop-shaping method produces less complex solutions for $C(s)$ than the brute force approach

☒ TRUE ☐ FALSE

Q3) $C(s)$ always follows the "conservation of dirt" law (Bode's Integral)

☐ TRUE ☒ FALSE

=> $S(s)$!!